

Radix Sort

Step 1 – Sort by Least Significant Digit (1s) Digits:

Number	Last Digit
564	4
213	3
987	7
432	2
123	3
765	5
321	1
654	4
876	6

After sorting by units place:
[321, 432, 213, 123, 564, 654, 765, 876, 987]

Step 2 – Sort by Tens Digit (10s) Tens digits:

Number	Tens Digit
321	2
432	3
213	1
123	2
564	6
654	5
765	6
876	7
987	8

After sorting by tens place:
[213, 321, 123, 432, 654, 564, 765, 876, 987]

Step 3 – Sort by Hundreds Digit (100s)

After sorting by hundreds place:
[123, 213, 321, 432, 564, 654, 765, 876, 987]

Python Implementation

```
def counting_sort(arr, exp):
    n = len(arr)
    output = [0] * n
    count = [0] * 10

    # Count occurrences of digits
    for i in range(n):
        index = (arr[i] // exp) % 10
        count[index] += 1

    # Change count[i] so that count[i] contains
    # the actual position in output
    for i in range(1, 10):
        count[i] += count[i - 1]

    # Build the output array
    i = n - 1
    while i >= 0:
        index = (arr[i] // exp) % 10
        output[count[index] - 1] = arr[i]
        count[index] -= 1
        i -= 1

    # Copy to original array
    for i in range(n):
        arr[i] = output[i]

    return arr
```

```
def radix_sort(arr):
    max_val = max(arr)
    exp = 1

    while max_val // exp > 0:
        arr = counting_sort(arr, exp)
        print(f'After sorting by digit {exp}: {arr}')
        exp *= 10

    return arr
```

```
numbers = [564, 213, 987, 432, 123, 765, 321, 654, 876]
```

```
sorted_numbers = radix_sort(numbers)
```

```
print("Final Sorted Output:", sorted_numbers)
```

3. Final Sorted Output

[123, 213, 321, 432, 564, 654, 765, 876, 987]

4. Reflection Questions

What is the order of sorting for digits in Radix Sort?
Digits are sorted from:

Least Significant Digit → Most Significant Digit
(1s → 10s → 100s ...)

This is called LSD Radix Sort.
After sorting by the least significant digit, what does the intermediate array look like?

[321, 432, 213, 123, 564, 654, 765, 876, 987]

After sorting by the most significant digit, what is the final sorted output?

[123, 213, 321, 432, 564, 654, 765, 876, 987]

5. Median and Order Statistics

Sorted list:

[123, 213, 321, 432, 564, 654, 765, 876, 987]

Number of elements:

$n = 9$ (odd)
Median formula:
Median = element at position $(n + 1) / 2$

Median position = $(9 + 1) / 2 = 5$
Median value: 564

6. Order Statistics from the Data

Statistic	Value
Minimum	123
Maximum	987
Median	564
3rd Order Statistic	321
7th Order Statistic	765